

Acute Decompensated Heart Failure — Inpatient Management

Profile the patient first (wet/dry × warm/cold) → decongest with IV loops, dose by home dose → start/continue GDMT before discharge · the admission is the intervention

1 · Profile on arrival — congestion & perfusion

Two bedside axes drive every decision. **Congestion** (wet vs dry) and **perfusion** (warm vs cold). Most floor patients are **warm-and-wet**.

	Dry	Wet (congested)
Warm (perfused)	Compensated — optimize GDMT	Warm-wet: IV loop diuretic — the bread & butter admit
Cold (hypoperfused)	Cold-dry: cautious fluid / inotrope	Cold-wet: diuresis + consider inotrope/vasodilator; think ICU

Congestion signs: JVD, orthopnea, edema, weight gain, B-lines/POCUS. **Hypoperfusion:** narrow pulse pressure, cool extremities, ↑lactate, ↑Cr, somnolence. Hypoperfusion ≠ hypotension.

2 · Decongest — IV loop diuretics

- **Give IV, not PO** — gut edema cripples PO absorption.
- **Dose:** IV furosemide at $\geq 1\text{--}2.5\times$ the total home daily PO dose, given IV BID (DOSE trial). Naive patient: 40–80 mg IV.
- **Check response at 2 h** (spot urine Na $> 50\text{--}70$ mmol/L or UOP $> 100\text{--}150$ mL/h = good). If inadequate, **double the dose** — don't just wait.
- Inadequate on high-dose loop → add **thiazide** (metolazone / IV chlorothiazide) for sequential nephron blockade; consider acetazolamide (ADVOR) or early SGLT2i.
- Target net negative $\sim 1\text{--}2$ L/day; daily weights, strict I/O, BMP q12–24h.

A rising creatinine during effective decongestion is often hemoconcentration, not injury — don't reflexively stop diuresis if the patient is still wet and making urine.

3 · GDMT — start it during admission

The four pillars of HFrEF ($EF \leq 40\%$). Initiate/continue **before discharge** — admission is the highest-yield moment (STRONG-HF).

Pillar	Agent	Inpatient note
ARNI/ ACEi/ARB	sacubitril-valsartan	Hold for SBP < 90 , K > 5.5 , AKI; 36 h washout from ACEi
Beta-blocker	carvedilol, metoprolol succ.	Continue if already on; don't start until euvolemic & off inotropes
MRA	spironolactone, eplerenone	K < 5.0 and eGFR > 30
SGLT2i	dapagliflozin, empagliflozin	Start in-hospital — safe, speeds decongestion (EMPULSE)

Don't stop a chronic beta-blocker just for an admission unless the patient is cold/hypoperfused — withdrawal worsens outcomes. Up-titrate GDMT at close follow-up.

Precipitants — find the FAILURE

- Forgetting meds / Fluid & salt · Arrhythmia (AF/RVR) · Ischemia/Infection · Lifestyle (Na, EtOH) · Upregulation (thyroid, pregnancy, anemia) · Renal/Rx (NSAIDs, gabapentinoids, dihydropyridines) · Embolus (PE)
- Work-up: ECG + troponin, BNP/NT-proBNP, CBC, BMP, TSH, CXR; echo if no recent EF.

Discharge bundle & Do/Avoid

- **DO** transition IV→PO diuretic & observe 24 h euvolemic before discharge
- **DO** document EF, dry weight, discharge diuretic dose, and the 4 GDMT pillars
- **DO** arrange **7-day** follow-up + daily-weight teaching
- **AVOID** discharging still-congested ('looks better') — residual congestion drives readmission
- **AVOID** NSAIDs, glitazones, most non-DHP CCBs in HFrEF
- **AVOID** starting/up-titrating beta-blocker while still wet or hypoperfused